

## Determine the Frequency of AC using Sonometer

**Aim:-** To find the frequency of alternating current (AC) by using sonometer and an electromagnet.

**Apparatus:**

- A sonometer with soft iron wire stretched over it
- An electromagnet with a step down transformer (220-6V)
- Slotted weights with hanger

**Principle:**

Let a wire of length  $l$  and mass per unit length  $m$  be stretched by applying a force  $T$  called the tension in the string. If it is set into transverse vibrations in fundamental mode, then the frequency  $n$

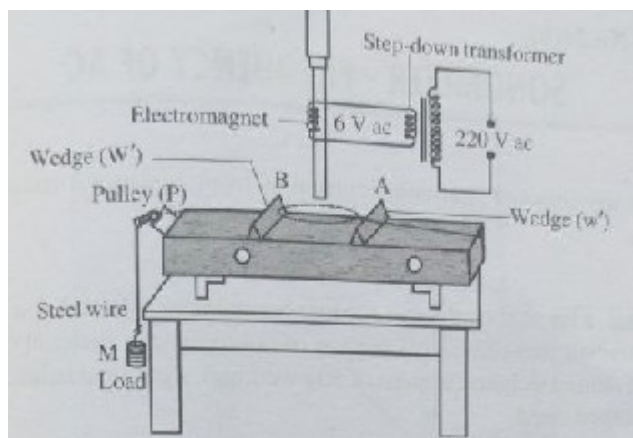
of the note emitted by it is given by 
$$n = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

Now if an alternating current is passed through an electromagnet, it will be temporarily magnetised twice in every cycle, firstly one of its faces becomes north pole and secondly the same face becomes south pole. If the sonometer wire is held with its middle point very close to the pole, it will be attracted towards the electromagnet, at the moment it is magnetised and thrown back when the electromagnetism is lost.

*In each cycle of AC, the wire is pulled and released twice because the steel wire is always attracted by the electromagnet, whether its face close to the wire becomes a south pole or a north pole.*

If the length  $l$  and the tension  $T$  is so adjusted that it is set into resonant vibrations by the AC, the frequency of AC will be half that of vibration of the wire. If  $n$  is the frequency of the sonometer wire thrown into resonant vibrations in fundamental mode,

then the frequency of AC is 
$$f = \frac{n}{2}$$



**Procedure:**

- Place the sonometer in a working table such that the pulley projects beyond the edge of the table.
- Stretch the sonometer wire by placing a load of 100g.
- Support the electromagnet in a stand and connect the electromagnet to the secondary of the (220V-6V) step down transformer. Adjust the position of the electromagnet in such a way that one of its poles lies close to the middle of the sonometer wire.
- Switch ON the AC supply; Adjust the length of the vibrating portion of the wire by sliding one of the knife edges (start with a small length between the two bridges) until the wire begins to vibrate. Now make minor adjustments until the amplitude of the vibrating wire segment is maximum. Measure the length  $l$  of the vibrating wire between the knife edges and note down the tension  $T$  (weight of hanger + weights).
- Now increase the load in steps of 100g weights and adjust the vibrating length by sliding one of the bridges such that the wire begins to vibrate with maximum amplitude in each case. In this way take five sets of observations for different loads.
- Find  $m$  the mass per unit length of wire.

**Observations:**

Mass per unit length of the wire =  $m =$

| S.No | Load in Kg<br>$M$ | Resonant Length in m<br>$l$ | Frequency in Hz<br>$n = \frac{1}{2l} \sqrt{\frac{Mg}{m}}$ |
|------|-------------------|-----------------------------|-----------------------------------------------------------|
|      |                   |                             |                                                           |
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|      |                   |                             |                                                           |
|      |                   |                             |                                                           |

Mean  $n = \dots\dots\dots$

Frequency of sonometer wire  $n = \dots\dots\dots$  Hz

$$\therefore \text{Frequency of AC } f = \frac{n}{2} = \dots\dots\dots \text{ Hz}$$

**Results:**

Frequency of AC =  $\dots\dots\dots$  Hz